Phase 15–16 Plan: D&O Risk Core and S&P 100

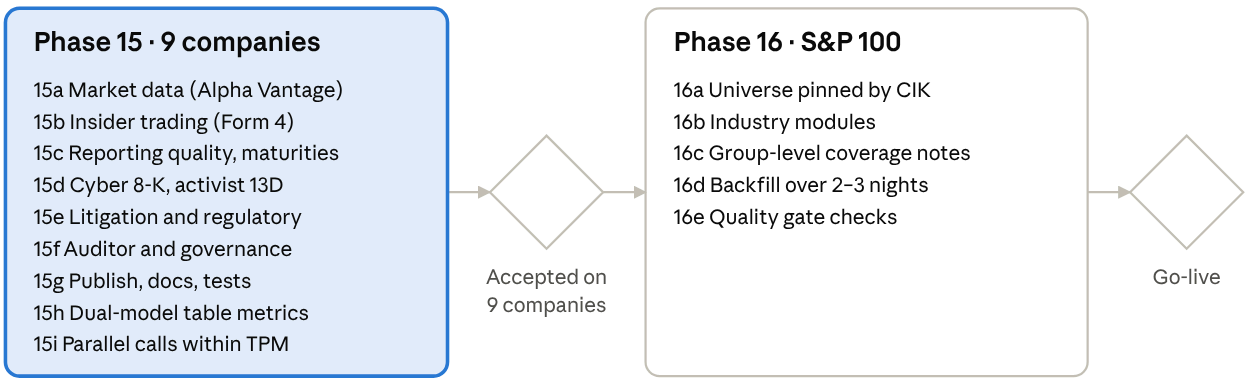
Oct 15, 2026 · @Jiehuan Huang

Summary

Phase 15 adds the drivers of D&O claims that the pipeline does not yet capture: stock drops, earnings misses, insider selling, reporting-quality red flags, and litigation and regulatory investigations. Phase 16 then extends the pipeline from 9 companies to the S&P 100.

- **Phase 15 (9 current companies):** market data, insider trading, reporting quality and debt maturities, new event signals, litigation and regulatory matters, auditor facts. The application database schema does not change.
- **Phase 16 (S&P 100):** a common core for every company, one metric module per industry, group-level coverage rules, and a staged first backfill.
- **Why in this order:** the new core is proven on 9 companies we already know well before it runs on 100.

Phase 16 starts only after Phase 15 passes on the 9 current companies



roadmap · 2 phases, 2 gates

Eight decisions are needed before work starts (last section). The Chinese working plan is in `docs/PLAN.md`.

Where we are today

The pipeline already covers financial health and governance well for 9 companies (MAR, LUV, F, PFE, JPM, M, LUMN, SKYD, ENPH). What it lacks is the market and disclosure side, which is where most D&O claims start.

Area	Covered today	Missing for D&O
Financial health	Revenue, margins, leverage, interest coverage, cash flow, buybacks (XBRL)	Debt maturities, Altman Z
Banks	ROA, ROE, CET1, NIM, NPL, loan book, reserve ratio	Uninsured deposits, unrealized securities losses, CRE concentration
Industry KPIs	RevPAR, occupancy, ADR (MAR); AUM, VaR (JPM); RPO	Most other industries
Events	8-K items (restatement, auditor change, departures, impairment, delisting, bankruptcy), going concern, material weakness	8-K 1.05 cybersecurity, activist 13D
Governance	Board independence, CEO-chair, lead director, pay versus performance	Auditor tenure and fees, dual-class shares, CEO tenure
Market	Nothing (Alpha Vantage is used only for news and transcripts)	Market cap, stock drops, volatility, earnings surprises
Insiders	Nothing	Form 4 insider selling
Litigation	Nothing	Securities class actions, SEC and DOJ investigations, product liability

A securities class action typically follows a stock drop after a disclosure, often with insider selling beforehand. None of those three are tracked today.

Phase 15: D&O risk core on the current 9 companies

Nine workstreams. Five need no LLM; litigation and part of the auditor work use the existing dual-model checks, 15h extends those checks to table metrics, and 15i runs model calls in parallel.

Workstream	Source	New metrics	New signals	LLM
15a Market data	Alpha Vantage `OVERVIEW` (weekly), `TIME_SERIES_DAILY` (daily), `EARNINGS` (after filings), SPY as benchmark	Market cap, 90-day volatility, 52-week max drawdown, EPS surprise % (quarter-end values)	Single-day drop of 10% or more and 8% or more below SPY; drop within 3 trading days of an earnings miss	No
15b Insider trading	SEC Form 4 (structured XML)	Quarterly net insider sales, share of officer holdings sold	3 or more officers or directors selling on the open market within 30 days, outside 10b5-1 plans; Warning when it precedes a drop or miss by up to 90 days	No
15c Reporting quality and solvency	XBRL: receivables, inventory, goodwill and impairment, stock compensation, diluted shares, debt maturities years 1–5, loss contingency accruals	DSO, DIO, accruals / assets, goodwill / equity, stock comp / revenue, diluted share change, debt due in 12 months / (cash + operating cash flow), Altman Z (non-financials)	DQ warnings: Altman Z below 1.81; near-term maturities above cash plus operating cash flow	No

Workstream	Source	New metrics	New signals	LLM
15d Event signals	8-K items; SC 13D / SCHEDULE 13D filed about the company	–	8-K 1.05 cybersecurity incident (Warning), 1.02 termination of a material agreement (Info), activist stake	No
15e Litigation and regulatory	10-K Item 3, contingencies note, 10-Q Part II Item 1	Matter list with type, agency, status, accrual and quote	New SEC or DOJ investigation, Wells notice, securities class action	Yes, both models must agree
15f Auditor and governance	10-K cover XBRL (`dei:AuditorName`); audit report text; proxy statement	Auditor, tenure, audit and non-audit fees, dual-class shares, CEO tenure	–	Proxy only, both models
15g Publish and docs	–	New categories "Market & Trading" and "Reporting Quality"; new risk types (Litigation, Regulatory, Market, Insider Trading, Cybersecurity)	–	No

How the riskiest pieces are kept honest:

- **Stock drops** are recomputed from stored daily prices, so every signal can be checked against the price series.

- **Litigation matters** are stored only when the quote is found in the filing and both models agree on the type. A matter that cannot be matched to an earlier one goes to the human review queue (`uw-pipeline review`).
- **Activist 13D filings:** we first confirm that EDGAR lists filings made about a company under that company. If it does not, a separate decision is recorded before using full-text search.

15h Dual-model check for displayed table metrics. MAR's RevPAR, occupancy and ADR are read from tables by one model today, protected by the cell check alone. They will need both models to pick cells that pass the check and agree within 0.5%, as the bank ratios already do. Table metrics used only as insight input (AUM, VaR, international revenue, average interest-earning assets, capacity utilization) stay single-model. Every displayed metric added in Phase 16 is dual-checked. Cost: one extra model call per filing that holds these tables. Acceptance: MAR's 5 stored periods do not change, and disagreements are counted in run statistics and listed in the log.

15i Parallel model calls, limited by TPM and RPM. Every model call runs one at a time today, so wall time is the sum of call latencies, not a quota limit: the 9-company cold run used about 3.2M tokens in 37.5 minutes, roughly 86K tokens a minute. 15i adds a token bucket per model deployment for tokens per minute (TPM) and requests per minute (RPM), with limits in `config/securegpt.yaml` confirmed by the gateway team. Each request reserves its estimated prompt tokens plus `max_tokens`, because Azure OpenAI counts that estimate against TPM. A 429 response backs off the whole bucket. Companies run in parallel within a stage, and independent calls inside a company run in parallel too (documents in text metrics, the two reads of a dual check, behavioral and strategic insights). Prerequisites, taken from Phase 10: model calls move outside database transactions, and concurrent cache writes keep the first result. `max_tokens` becomes per task, so oversized limits do not use up quota. SQLite development stays at one worker. Acceptance: the 9-company cold run drops from 37.5 to under 10 minutes, with no failures from 429s and the same results as a serial run.

Phase 16: extend to the S&P 100

At 100 companies nothing is tuned per company: every company gets the common core, plus the module for its industry.

16a Universe. Each company is pinned by CIK from the S&P 100 list as of a set date. Share classes of one company count once (GOOGL and GOOG). Special symbols such as BRK.B get an Alpha Vantage mapping. The list is reviewed quarterly. A company-level industry

override is added for cases where the SIC code is too coarse (AMZN files under 5961, catalog retail).

16b Industry modules. Financial companies get their own core (ROA, ROE, leverage, regulatory capital) in place of current ratio and EBITDA. Each group is XBRL-first, with at most 3–5 text or table metrics, and each such metric is tested on at least 2 holdout companies.

Group	S&P 100 examples	Module metrics
Software and internet	MSFT, ORCL, CRM, ADBE, GOOGL, META	RPO, deferred revenue growth, R&D / revenue, stock comp / revenue
Semiconductors and hardware	NVDA, AVGO, AMD, INTC, AAPL, QCOM	Gross margin, DIO, customer concentration (text)
Pharma and biotech	LLY, MRK, ABBV, PFE, AMGN, GILD, BMY	R&D / revenue, top-3 product share, patent expiries and product liability accruals (text)
Medtech and managed care	TMO, ABT, DHR, MDT, ISRG, UNH, CVS	Gross margin; medical loss ratio and membership (text)
Banks	JPM, BAC, WFC, C, USB	Today's bank set, plus uninsured deposits (text), deposit change, unrealized securities losses, CRE concentration
Broker-dealers and asset managers	GS, MS, SCHW, BLK, BK	AUM and net flows, VaR (table reading), fee share, leverage
Payments and consumer finance	V, MA, AXP, PYPL, COF	Payment volume (text), charge-offs, reserve coverage
Insurance	BRK.B, MET, AIG	Combined ratio (text), reserve development, investment yield
Retail and consumer	AMZN, WMT, COST, HD, LOW, TGT, MCD, PG, KO, PEP	Comparable sales (text), gross margin, DIO
Energy and utilities	XOM, CVX, COP, NEE, DUK, SO	Production, reserves; regulated asset base, disaster accruals (text)

Group	S&P 100 examples	Module metrics
Industrials, aerospace and defense	CAT, DE, HON, GE, RTX, LMT, BA, GD	Backlog (RPO), warranty reserves, pension deficit
Transport	UNP, UPS, FDX	Operating ratio, unit costs
Telecom and media	T, VZ, TMUS, CMCSA, DIS, NFLX, CHTR	Leverage, ARPU and net adds (text), content amortization
Autos	TSLA, F, GM	Deliveries (press release), warranty and recall reserves, captive finance
REITs	AMT, SPG	FFO (press release), occupancy, debt maturities

16c Coverage rules. Coverage notes can be written per industry group ("current ratio does not apply to banks"). Company notes are kept for real exceptions. A gap with no note at any level becomes a DQ Warning.

16d Operations. The first backfill runs in 2–3 nightly batches by industry. After that, the nightly run processes only companies with new filings. If the nightly run outgrows its window, two items are pulled forward from the 4,000-company plan (Phase 10): shorter database transactions and parallel lanes.

16e Quality gate before go-live. Golden values from the filings for 2 companies per group. `audit-insights` on 20 companies. No DQ Errors. No unexplained coverage gaps. A stability check on 5 companies.

Data model changes

Most new metrics need no schema change. `metric_value` (pipeline) and `MatrixData` (application) hold one row per company, metric and period, so a new metric is new rows. New industry groups are config plus rows in `metrics`, `metric_category` and `SIC_Preferences`. New signals use the existing `signal` table and reach `StrategicRisk` through the insights stage.

Schema changes are needed only for data that is not quarterly or annual.

Pipeline database (our own tables; the engineer reruns `deploy/sql/pipeline_schema_full.sql`, which creates only the missing tables):

Change	Why
New `market_daily` (company, trade date, close, volume, shares outstanding)	Daily prices do not fit FY/Q rows; volatility, drops and drawdowns are computed from it
New `insider_transaction` (filer, role, trade date, code, shares, price, shares after, 10b5-1 flag, accession)	Form 4 is per transaction; details must be kept to aggregate and to audit
New `legal_matter` (type, agency, status, first disclosed, last seen, accrual, quote, document)	A matter is an item tracked across filings, not a number
New `audit_facts`, or columns on `board_composition` (chosen during build)	Auditor, tenure, audit and non-audit fees
`board_composition`: dual-class, CEO tenure	Governance metrics
`company`: industry sector, index membership	Grouping and filtering 100 companies
`metric_value.derivation`: new value MARKET	Separates market data from filing data

Application database (read by the BFF): two options.

	Option A (recommended first)	Option B
Schema change	None	New tables, for example `StockPrice`, a litigation list, insider transactions, a governance panel
How data shows	Market, insider and quality metrics in `MatrixData` at quarter end; drops, insider clusters and lawsuits as `StrategicRisk` items with quotes and links	Dedicated views: price chart with drop markers, matter list, transaction table, board panel
Who must change	Pipeline only	Pipeline, BFF and front end
Limit	No daily price chart, no matter or transaction list	More work and coordination

Board composition is not published to the application database today; it reaches users only through insights. Under Option A that stays the same.

Scale: `MatrixData` holds about 12,800 rows for 9 companies and will hold roughly 200,000 or more for 100. That is fine for SQL Server once `(CompanyId, metric_id)` is confirmed to be indexed.

Cost, quotas and run time

We use the paid Alpha Vantage key (75 calls a minute, no daily cap). A full cold run of 9 companies took about 38 minutes and $6 of LLM spend on 2026-10-14; the 100-company figures below scale from that.

Item	Phase 15 (9 companies)	Phase 16 (100 companies)
Alpha Vantage calls per day	About 30; paid key allows 75 a minute	About 300; under 5 minutes at 75 a minute
SEC requests	Small; Form 4 adds a few per company	One companyfacts call per company per day; fine
LLM, first full run	Litigation and proxy additions, a few dollars	About $70–100
LLM, nightly	Only companies with new filings	Same; peaks in earnings season
Daily LLM budget (`LLM_DAILY_BUDGET`)	$25 (current)	$40, raised temporarily in earnings season
Run time	Current single process	First backfill about 7 hours serial, 1–2 hours with parallel calls (15i); nightly runs incremental

All estimates assume the Claude profile; costs on SecureGPT have not been measured yet.

LLM cost and process time by metric group. Per company, cold run, serial. Current rows come from the 2026-10-14 run ($6 and 37.5 minutes for 9 companies) and run 37 stage timings; planned rows are estimates until Phase 15 records per-stage cost in run statistics.

Metric or insight group	Status	LLM	LLM cost per company	Process time per company
XBRL financial health and bank metrics	Current	No	$0	Under 5 s (one companyfacts call)
Industry table and text metrics, governance, event signals	Current	Yes	About $0.33	About 63 s (JPM 335 s)
Insights (behavioral and strategic)	Current	Yes	About $0.33	About 133 s
SEC fetch, publish, DQ	Current	No	$0	About 50 s
Current total	Current		**About $0.67**	**About 4.2 min**
15a Market data	Phase 15	No	$0	Seconds (about 3 Alpha Vantage calls)
15b Insider trading (Form 4)	Phase 15	No	$0	Seconds (a few SEC requests)
15c Reporting quality, Altman Z	Phase 15	No	$0	Under 1 s (reuses companyfacts)
15d Event signals (8-K 1.05, 1.02, 13D)	Phase 15	No	$0	Seconds
15e Litigation and regulatory	Phase 15	Yes, both models	Est. $0.15–0.30	Est. 60–120 s
15f Auditor and proxy governance	Phase 15	Proxy only, both models	Est. $0.05–0.15	Est. 30–60 s
15h Dual check on displayed table metrics	Phase 15	One extra call per filing with these tables	Est. under $0.05 (MAR only today)	Est. 10–20 s
16b Industry module text and table metrics	Phase 16	Yes, dual-checked	Est. $0.10–0.30	Est. 30–90 s

Metric or insight group	Status	LLM	LLM cost per company	Process time per company
Planned total	Phase 16		**Est. $0.70–1.00**	**Est. 5–8 min serial; 1–2 min with 15i**

The planned total matches the $70–100 first-run estimate for 100 companies. Rule-only rows add no LLM cost; 15i cuts wall time, not cost.

Where the time goes at several hundred companies a day. The first bottleneck is serial model calls; once 15i runs them in parallel, the bottleneck becomes the SecureGPT TPM and RPM quota. Estimate for 500 companies, from per-company figures in run 37:

Stage	Runs for	Serial time	Load	Bottleneck?
insights	Most companies daily (new news changes the input; the `news_materiality` gate removes some, to be measured)	About 133 s per company, 18.5 h	About 150K input + 25K output tokens per company; 75M + 12M a day	Yes: needs about 7 concurrent synthesis calls and about 500K TPM for a 3-hour window
textmetrics, governance, signals	Companies with new filings: 100–150 a day in earnings season	About 63 s per company (JPM 335 s), 2.6 h	Mostly the extractor model	Yes, in earnings season
SEC fetch	All companies	Under 10 min	A few thousand requests at 8 per second	No
Alpha Vantage news	All companies, 4 times a day	About 7 min per round	500 calls at 75 per minute (paid plan)	No
publish, dq	All companies	About 8 min	Azure SQL (SQLite allows one writer)	No

Acceptance criteria

Each phase is accepted only when every check below passes.

Phase 15

- Ford's single-day drop of about 18% on 2024-07-25, after its Q2 2024 earnings miss, is detected as both a stock drop and an earnings-miss drop.
- Every drop signal for the 9 companies over the last two years can be traced in the stored price series.
- Form 4 transaction counts match the EDGAR listings, and 10b5-1 flags are correct.
- New XBRL metrics match golden values taken from the filings, for at least 2 companies per metric type, with zero difference.
- Litigation: in a sample of 20 matters, every quote is found in its filing and at least 90% have the correct type.
- Rule-only stages make no LLM calls, and new LLM cost appears in run statistics.
- No DQ Errors; the full test suite passes on SQLite and SQL Server.

Phase 16

- All 100 companies run end to end.
- Every common-core metric has a value or a group or company note for each company.
- Golden values pass for 2 companies in each industry group.
- First-backfill cost and duration are recorded in run statistics.
- The nightly incremental run finishes inside its window.

Decisions needed

#	Decision	Options	Recommendation
1	Insider trading source	SEC Form 4, or Alpha Vantage `INSIDER_TRANSACTIONS`	SEC Form 4: authoritative, free, keyed by CIK, carries the 10b5-1 flag
2	Stock drop threshold	10% single-day drop and 8% or more below SPY; or absolute drop only	Both conditions, so market-wide sell-offs are not flagged

#	Decision	Options	Recommendation
3	Application database	Option A (no schema change) or Option B (new tables, BFF and front end)	Option A first; Option B only when product asks for the new views
4	Alpha Vantage plan	Paid plan from Phase 15	Decided: paid key in use, 75 calls a minute
5	Litigation timing	In Phase 15, or after Phase 16	Decided: In Phase 15: it is the largest D&O gap and is easier to validate on 9 companies
6	Coverage notes	Company level only, or industry-group level as well	Group level, with company notes for real exceptions
7	S&P 100 list	Reference date and owner of the quarterly review	To be named
8	SecureGPT quota	TPM and RPM per deployment (GPT-6 Luna for routine calls, GPT-6 Sol for tiebreakers) in dv, pp and pr, and any APIM subscription limit	Ask the gateway team; 15i needs the numbers, and about 500K TPM covers 500 companies a day